Shikhar Prasad

EDUCATION

Indian Institute of Technology Varanasi (IIT-BHU) 2018 - 2023
Bachelor's and Master's Degree (IDD- B.Tech + M.Tech), Electrical Engineering

SKILLS

Technologies: Django, Docker, Redis, Celery, AWS, NodeJs, Flask, Linux, Git, CI/CD, Distributed Systems
Languages: Python, JavaScript, C++, Java
Databases: PostgreSQL, Redis, MySQL, MongoDB

WORK EXPERIENCE

ARAMYA | Software Development Engineer — Remote | August 2024 – Present

- Monitored query performance using pg_stat_statements and integrated slow-query tracing with OpenTelemetry, reducing 99th percentile latency on critical endpoints and improving DB observability
- Developed and architected a video processing pipeline using Django, AWS MediaConvert and Celery with automated retries, real-time job tracking and multi-resolution output and reducing operational costs by 70%
- Built an intelligent ETL pipeline with dynamic schema detection, data type migrations that processes CSV files from AWS S3 into PostgreSQL tables, improving analytics efficiency and eliminating manual data workflows
- Implemented custom rate-limiting middleware to safeguard internal APIs from misuse and manage backend load during peak traffic periods

AEREM | Software Development Engineer — Remote | July 2023 – July 2024

- Implemented REST APIs in Node.js managing request, response, error codes, and failure call flows with reduced latency by leveraging indexing and asynchronous processing, resulting in overall efficiency gain
- Developed and optimized database schemas for efficient data storage and retrieval, enabling accurate calculation of key variables for enhanced analytics and decision-making, building analytical algorithms in Flask
- Built push notification service and implemented multi-user authentication and support for secure access

VORROCONNECT | Software Engineering Intern — Remote | June 2022 – August 2022

- Worked on the main Java-based service code, which aggregated diverse medical data from clients and transformed it into a universal structured format for client insights
- Debugged a production issue causing increased API latency on the Linux server

FREECHARGE | Product Manager Intern — Remote | May 2022 – June 2022

- Worked in sync with a team of developers to build a data system that extracted valuable insights from texts which was used for data analysis
- Researched user behavior and led strategic product roadmap discussion

PROJECTS

DISTRIBUTED LOCK MANAGEMENT UTILITY — Redis, Python, Django

- Implemented a Redis based distributed lock utility with context manager pattern and configurable timeout controls which was used in shipment status updates, order fulfillment pipelines, and authentication workflows
- Ensured atomic operations in high-concurrency scenarios and is now used extensively at Aramya

SORTING VISUALIZER — React, JavaScript

- Built a React Application to visualize various sorting algorithms, bootstrapping the project and using a component-based approach
- Implemented Merge, Quick, Bubble, Selection, and Insertion sort algorithms

COURSES COMPLETED

- **Core:** C++ Programming, Algorithms, Data Structures, Graph Theory, Artificial Intelligence
- **Mathematics:** Mathematical Methods, Probability and Statistics
- **Specialized:** Machine Translation, Modern Control Theory, Financial Markets